

- You may refer to paper notes and textbooks to complete this Quiz.
- You may not use the internet or access mobile phones or computers.
- You have 15 minutes to complete this quiz. You may have additional time you have Blundon accommodations for extra time
- You may use a calculator. If you do not have one, you may ask the instructor to complete a calculation

1. Complete the table below using the following data. (Formula and t-values needed to complete these calculations will be provided on the chalkboard. You will not be penalized for rounding or leaving your answer as a fraction or with a square root. Show your work) [6 marks]

| Blood CO <sub>2</sub><br>concentration<br>(mm Hg) |     |
|---------------------------------------------------|-----|
| 1                                                 | 50  |
| 2                                                 | 50  |
| 3                                                 | 50  |
| 4                                                 | 50  |
| 5                                                 | 50  |
| 6                                                 | 50  |
| 7                                                 | 60  |
| 8                                                 | 60  |
| 9                                                 | 80  |
| 10                                                | 100 |

| $y_i - \bar{y}$ | $(y_i - \bar{y})^2$ |
|-----------------|---------------------|
| -10             | 100                 |
| -10             | 100                 |
| -10             | 100                 |
| -10             | 100                 |
| -10             | 100                 |
| -10             | 100                 |
| 0               | 0                   |
| 0               | 0                   |
| 20              | 400                 |
| 40              | 1600                |
|                 | <u>2600</u>         |

$$\bar{y} = \frac{600}{10} = 60$$

$$SD = \sqrt{\frac{2600}{10-1}} = 17.0$$

$$SE = \frac{SD}{\sqrt{n}} = \frac{17}{\sqrt{10}} = 5.4$$

$$t_{0.025,9} = -2.26$$

$$t_{0.025,9} SE = (-2.26) 5.4 = 12.2$$

|                                        | n  | Mean<br>(mm Hg) | Median<br>(mm Hg) | SD<br>(mm Hg) | Variance<br>(mm Hg) <sup>2</sup> | SE<br>(mm Hg) | 95% CI for mean<br>(mm Hg) |
|----------------------------------------|----|-----------------|-------------------|---------------|----------------------------------|---------------|----------------------------|
| Blood CO <sub>2</sub><br>concentration | 10 | 60              | 50                | 17.0          | 288.9                            | 5.4           | [47.8,<br>72.2]            |

2. What is the meaning of the 95% confidence interval? [2 marks]

There is a 95% chance that this interval contains the population mean (or parameter value of interest).

3a. What is the name of the interval that represents the interpretation of the population mean as a random variable. For example, this interval can be interpreted as the population mean having a 95% chance of being within this interval? [1 mark]

a particular value

credible interval

3b. What is type of statistical inference is used to estimate this interval? [1 mark]

Bayesian.

4a. Historically, hypotheses have been tested with a Type I error rate of  $\alpha = 0.05$ . What does this mean? [2 marks]

Type I error is the probability our test statistic is above the critical value when the null hypothesis is correct.

No effect occurs, but effect claimed.

4b. Name a disadvantage of using a lower Type I error threshold, i.e.,  $\alpha = 0.01$ ? Full explain any new terminology you introduce in answering this question. [2 marks]

- Large sample size needed to detect effect.

- Increases chance of Type II error. Type II error is the probability of failing to detect an effect that actually occurs.

everyone was awarded full marks. due to typo. ↓ will be re-tested.